TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the Surface Tension of Water by the Capillary Tube Method

Water climbs a fine tube until surface tension's pull balances the column's weight — measuring the climb and the bore gives T = ½ρgr(h + r/3).

PHY-1104 · PHYSICS-I SESSIONAL · EXPERIMENT 6 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to find the surface tension of water from capillary rise. In a tube of radius r, the surface-tension force around the meniscus's rim (2πrT, pulling at the contact angle θ ≈ 0 for water on clean glass) supports the weight of the raised column. Including the small correction for the liquid held in the meniscus itself, the working formula is

$$T = \tfrac{1}{2}\,\rho\, g\, r\,(\,h + r/3\,)$$

where h is the rise from the reservoir's free surface to the bottom of the meniscus and ρ the liquid's density (Physics-I Topic 3 carries the full theory). The height is read with a travelling microscope; the bore r is found afterwards by measuring a mercury thread in the tube.

2. Apparatus and procedure

- **Apparatus:** capillary tubes of several bores, a beaker of distilled water on an adjustable stand, travelling microscope, a pin/pointer for the reservoir level, mercury and scale for the bore measurement.
- Clean the tube thoroughly (the experiment fails on grease): wash with chromic acid or detergent, rinse with distilled water, dry.
- Mount the tube vertically in the water; let the rise settle. Read the reservoir surface level (microscope on the tip of a pointer just touching the surface) and the meniscus bottom; h is the difference.
- Repeat for tubes of different bores: h should scale as 1/r — the experiment's built-in check.
- Measure each bore: draw a short mercury thread into the tube, measure its length l and mass m; r = √(m/(πlρ_Hg)).

3. Worked calculation (model data)

RESULT

Tube bore from the mercury thread: thread length 5.00 cm, mass 0.534 g, $\rho_{Hg} = 13{,}600$ kg m^{-3} → volume = $5.34 \times 10^{-4}/13{,}600\ldots$ (in SI: $m = 5.34 \times 10^{-4}$ kg; $V = m/\rho = 3.93 \times 10^{-8}$ m^3; $r = \sqrt{(V/(\pi l))} = \sqrt{(3.93 \times 10^{-8}/(\pi \times 0.0500))} = \sqrt{(2.50 \times 10^{-7})} = 5.0 \times 10^{-4}$ m — a 0.50 mm radius.
Rise: $h = 2.94$ cm $= 0.0294$ m. Water: $\rho = 1000$ kg m^{-3}.
$T = \tfrac{1}{2} \times 1000 \times 9.8 \times 5.0 \times 10^{-4} \times (0.0294 + 0.0005/3) = 2.45 \times (0.02957) =$ **0.0724 N m^{-1}** — water's accepted surface tension near room temperature (0.0728 N m^{-1} at 20 °C).

4. Precautions

- Cleanliness decides this experiment: a trace of grease raises the contact angle and lowers the rise. Fresh distilled water, freshly cleaned glass.
- The tube must be vertical and the rise allowed to settle (approach from above the final level — let the column fall to its height, since a rising meniscus sticks).
- Read the **bottom** of the (concave) meniscus, with the microscope levelled and no parallax.
- Record the water temperature: T falls as temperature rises, and the accepted value quoted for comparison must match the temperature.
- Use several bores and check $h \propto 1/r$; one bad bore reading then stands out instead of silently poisoning a single-tube result.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Derive $T = \tfrac{1}{2}\rho g r h$ from a force balance (neglecting the meniscus correction), stating where the contact angle enters.
2. A tube of radius 0.35 mm gives a rise of 4.20 cm for water. Compute T (take $\rho = 1000$ kg m^{-3}, neglect r/3).
3. Why is the reservoir level read with a pointer tip rather than by eye at the beaker's side?
4. What does the r/3 term correct, and when can it be dropped?
5. A tube gives a much lower rise than its neighbours of similar bore. Give the most likely cause and the remedy.
6. Why does the rise scale as 1/r, and how is that used as a check?

Answer key

1. The meniscus rim has length $2\pi r$; surface tension pulls along it at angle θ to the vertical, giving an upward force $2\pi r T\cos\theta$. This supports the column's weight $\pi r^2 h\rho g$. Equating: $2\pi r T\cos\theta = \pi r^2 h\rho g \rightarrow T = \rho g r h/(2\cos\theta)$. For water on clean glass $\theta \approx 0$ and $\cos\theta = 1$, giving $T = \tfrac{1}{2}\rho g r h$; the r/3 term then accounts for the liquid in the meniscus cap itself.
2. $T = \tfrac{1}{2} \times 1000 \times 9.8 \times 3.5 \times 10^{-4} \times 0.0420 = 4900 \times 3.5 \times 10^{-4} \times 0.0420 = 4900 \times 1.47 \times 10^{-5} = 0.0720\ \text{N m}^{-1}$ ✓ consistent.
3. The beaker's wall distorts the surface (its own meniscus) and the eye cannot judge the flat central level through curved glass. A pointer whose tip just touches the undisturbed central surface gives a point the microscope can be set on exactly, before and after — the difference of two microscope readings is then a true height difference.
4. It adds the weight of the liquid contained in the curved meniscus region above the column's flat top — approximated as a cylinder of height r/3. It matters only when r is not negligible beside h; for very fine tubes ($r \ll h$) it is dropped.
5. Contamination — grease or dirt in that tube has raised the contact angle ($\cos\theta < 1$) and spoiled the wetting. Remedy: re-clean the tube rigorously (chromic acid/detergent, distilled-water rinse) and repeat; the check of Q6 is what exposes the bad tube.
6. Because the supporting force grows with the rim ($\propto r$) while the column's weight grows with its cross-section ($\propto r^2$): $h = 2T/(\rho g r) \propto 1/r$. Plotting h against 1/r for several tubes must give a straight line through the origin — a bent point betrays a mis-measured bore or a dirty tube.

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